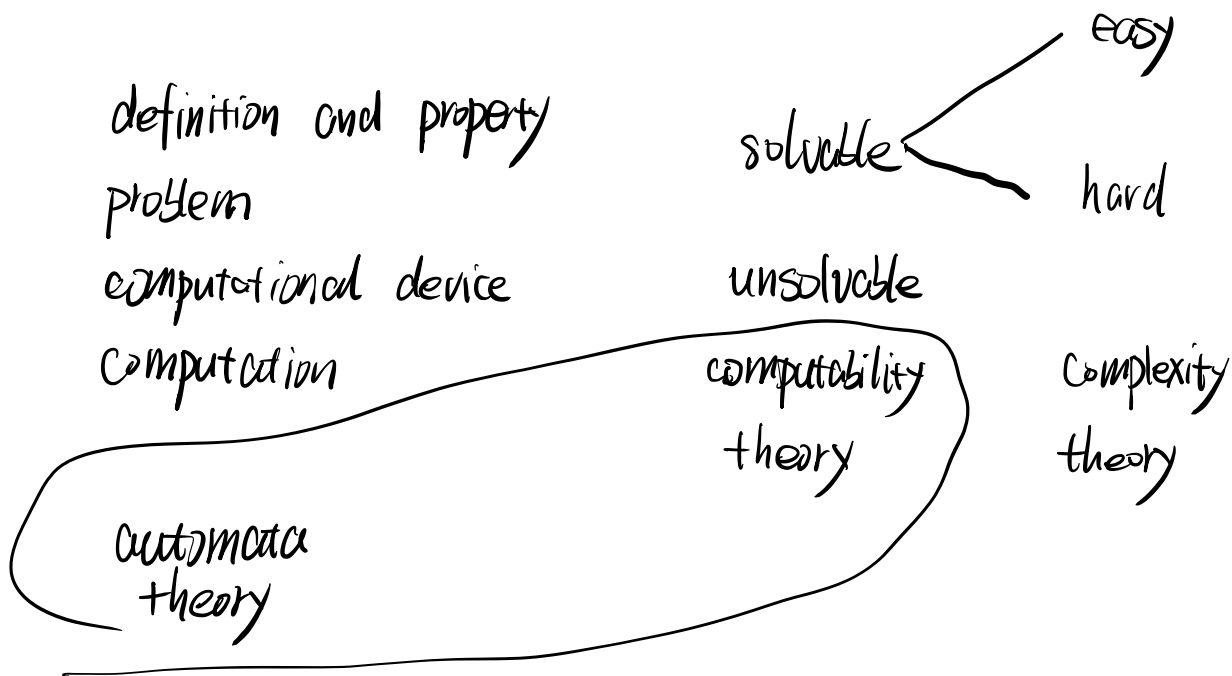


# Theory of Computation

can do | cannot do



## Problems and languages

An alphabet is a <sup>\*</sup>finite<sup>\*</sup> set of symbols.

$$\Sigma = \{0, 1\}$$

$$\Sigma = \{a, b, c, \dots, x, y, z\}$$

$$\Sigma = \{0, 1, 0\}$$

$$\Sigma = \{ \}$$

symbol

00 010

a, ab, abc, cba

0, 010

over

A string is a <sup>\*</sup>finite<sup>\*</sup> sequence of symbols from some alphabet  $\Sigma$ .

01 over  $\{0, 1\}$

01 over  $\{0, 1, 2\}$

length of a string  $w = \# \text{ symbols in } w \quad |w|$

If  $|w|=0$ , the empty string,  $\epsilon$

$\Sigma^i$  = the set of all strings of length  $i$  over  $\Sigma$ .

$$\Sigma = \{0, 1\}$$

$$\Sigma^0 = \{\epsilon\}$$

$$\Sigma^1 = \{0, 1\}$$

$$\Sigma^2 = \{00, 01, 10, 11\}$$

-----

$\Sigma^+$  = the set of <sup>all</sup> non-empty strings over  $\Sigma$ .

$$\Sigma^+ = \Sigma^1 \cup \Sigma^2 \cup \Sigma^3 \cup \dots$$

$\Sigma^*$  = the set of all strings over  $\Sigma$ .

$$\Sigma^* = \Sigma^+ \cup \{\epsilon\}$$

concatenate

$$vw$$

$$v = 010$$

$$w = 100$$

$$vw = 010100$$

string exponentiation

$$w^i = \underbrace{w \cdot \dots \cdot w}_i$$

$$w = 01$$

$$w^0 = \epsilon$$

$$w^1 = 01$$

$$w^2 = 0101$$

$$w^3 = 010101$$

$$w^0 = \epsilon$$

$$w^{i+1} = w \cdot w^i$$

definition by induction

Reversal

$$w = a_1 a_2 \dots a_n$$

$$w^R = a_n a_{n-1} \dots a_1$$

$$w = 0101 \quad w^R = 1010$$


---

If  $|w| = 0$ ,  $w^R = e$

If  $|w| \geq 1$ ,  $w = ua$  for some  $u \in \Sigma^*$ ,  $a \in \Sigma$

$$w^R = a u^R$$

A set of strings over some alphabet  $\Sigma$  is called a language.

$$L_1 = \{0, 10, 100, 1000, \dots\} \text{ over } \{0, 1\}$$

$$\Sigma = \{0, 1\} \quad L_2 = \{1, 11, 101, \dots\} \text{ over } \{0, 1, 2\}$$

$$L_3 = \emptyset$$

$$L_4 = \Sigma^*$$

$$L_5 = \{0^n 1^n : n \geq 0\}$$

language

$L$  over  $\Sigma$



(decision)

problem

Given  $x \in \Sigma^*$ ,  
is  $x \in L$ ?

$$L = \{ \text{encode}(G) : \left. \begin{array}{l} G \text{ is} \\ \text{tree} \end{array} \right\}$$

over  $\{0, 1\}$

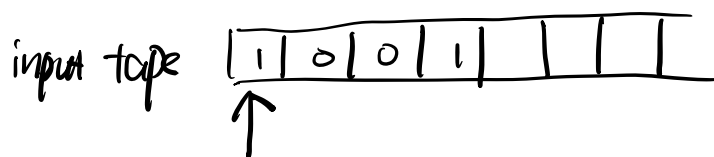
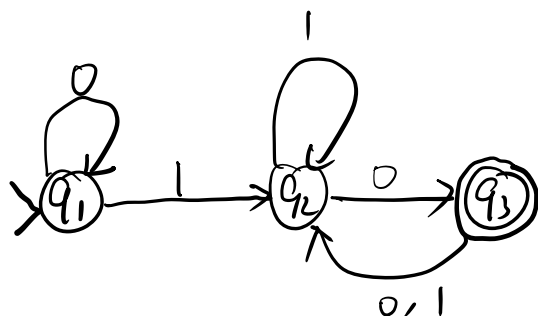
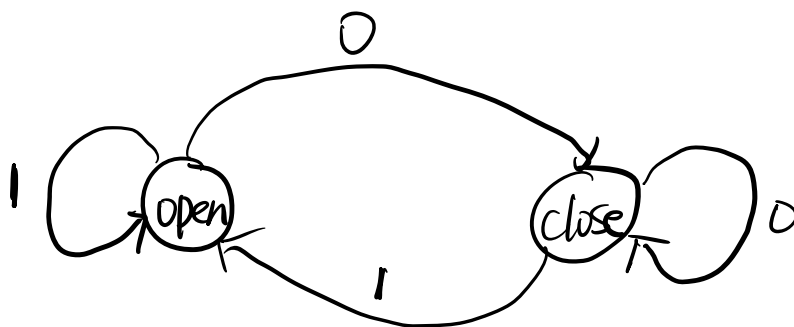
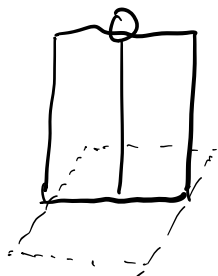
Given a graph  $G$ ,  
is  $G$  a tree?

Given  $x \in \{0, 1\}^*$ ,  
is  $x \in L$ .



# Computational device and computation

## finite automata



$q_1 \xrightarrow{1} q_2 \xrightarrow{0} q_3 \xrightarrow{0} q_2 \xrightarrow{1} q_2$

If ends with a final state  
accept input.

$\circ$ : state

$\rightarrow \circ$ : initial state (unique)

$\odot$ : final state (several)

state diagram

A <sup>deterministic</sup> (DFA) finite automata is 5-tuple  $(K, \Sigma, \delta, s, F)$ :

- $\circ K$ : a <sup>\*</sup>finite set of states

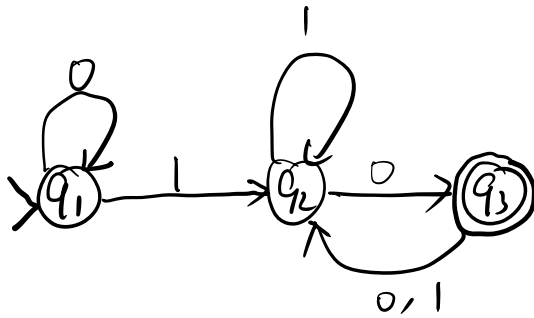
- $\circ \Sigma$ : a finite set of input symbols.

- $\circ s \in K$ : initial state

- $\circ F \subseteq K$ : a set of final states.

•  $\delta: K \times \Sigma \rightarrow K$  transition function

$\downarrow$  current state     $\downarrow$  symbol     $\downarrow$  next state



$$K = \{q_1, q_2, q_3\}$$

$$\Sigma = \{0, 1\}$$

$$s = q_1$$

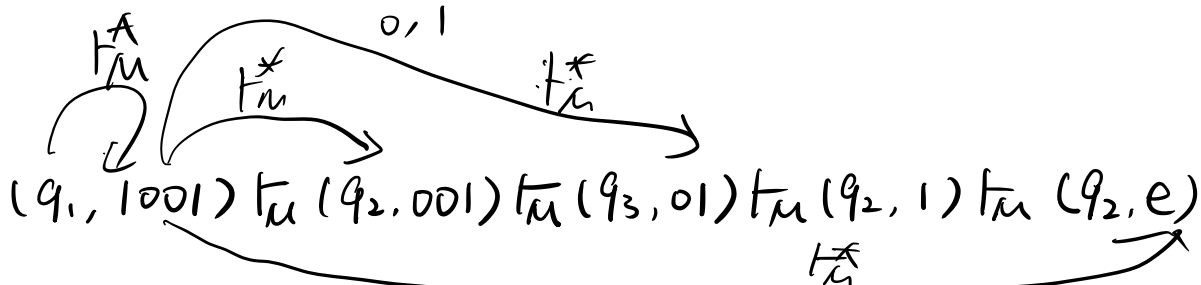
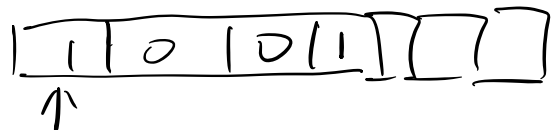
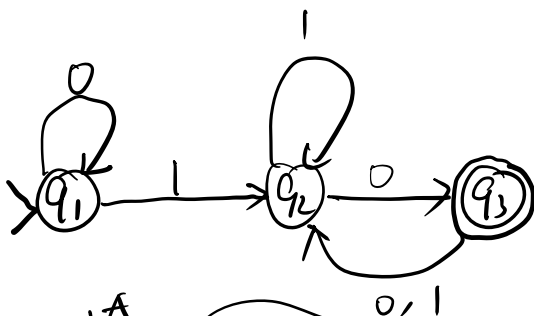
$$F = \{q_3\}$$

$\delta:$

| $q$   | $a$ | $\delta(q, a)$ |
|-------|-----|----------------|
| $q_1$ | 0   | $q_1$          |
| $q_1$ | 1   | $q_2$          |
| $q_2$ | 0   | $q_3$          |
| $q_2$ | 1   | $q_2$          |
| $q_3$ | 0   | $q_2$          |
| $q_3$ | 1   | $q_2$          |

A configuration of a finite automata  $M = (K, \Sigma, \delta, s, F)$  is an element of  $K \times \Sigma^*$ .

$\downarrow$  current state     $\downarrow$  unread input



$(q, w) \vdash_M (q', w')$  if  $w = aw'$  for some  $a \in \Sigma$   
 and  $\delta(q, a) = q'$ .  
 $\downarrow$   
 yields in one step

$(q, w) \vdash_M^* (q', w')$  if  $(q, w) = (q', w')$  or  
 $\downarrow$   
 yields  
 $\exists (q_0, w_0), \dots, (q_i, w_i)$  for  $i \geq 1$   
 st  
 $(q, w) = (q_0, w_0) \vdash_M \dots \vdash_M \dots \vdash_M (q_i, w_i)$   
 $\parallel$   
 $(q', w')$

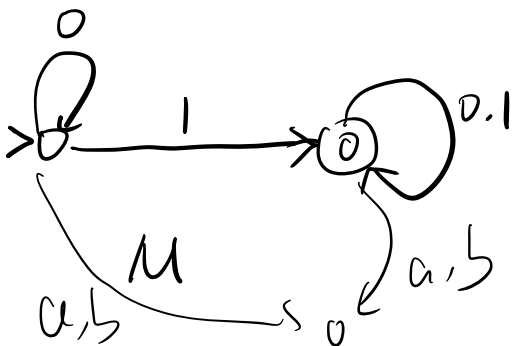
$M$  accept  $w \in \Sigma^*$  if  $(s, w) \vdash_M^* (q, \epsilon)$  for some  $q \in F$ .

$L(M) = \{ w \in \Sigma^* : M \text{ accepts } w \}$

$M$  accepts/reorganize  $L(M)$ .

$M$  accepts  $L$ 

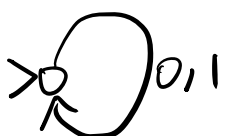
1.  $L$  contains every string accepted by  $M$ .
2. every string in  $L$  is accepted by  $M$ .



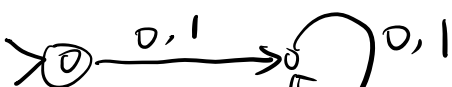
1.  $M$  accepts  $01$   $\top$

2.  $M$  accepts  $\{01\}$   $F$

$M$  accepts  $1$  but  $1 \notin \{01\}$ .



$L(M) = \{ \}$

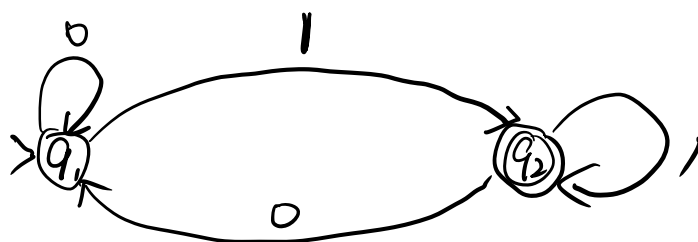


$L(M) = \{ \epsilon \}$

e { }

A language is regular if it is accepted by some DFA.

Exercise show  $\{w \in \{0,1\}^* : w \text{ ends with } 1\}$  is regular.



$q_1$ : the last digit we read is not 1

$q_2$ : " " is 1

## Regular Operation

Union  $A \cup B = \{w : w \in A \text{ or } w \in B\}$

Concatenation  $A \circ B = \{ab : a \in A \text{ and } b \in B\}$

$A = \{\text{good}, \text{bad}\}$

$B = \{\text{cat}, \text{dog}\}$

$A \circ B = \left\{ \begin{array}{l} \text{goodcat}, \text{gooddog}, \\ \text{badcat}, \text{baddog} \end{array} \right\}$

Star  $A^* = \{w_1 w_2 \dots w_k : w_i \in A \text{ and } k \geq 0\}$

$\epsilon \in A^*$

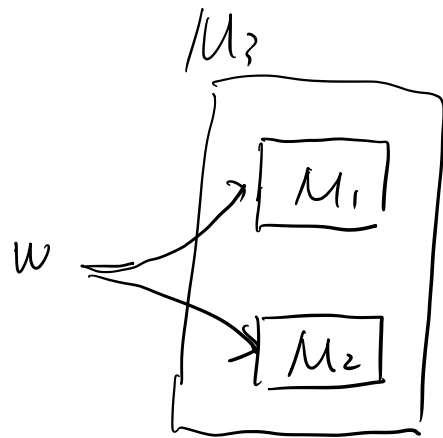
Theorem

If  $A$  and  $B$  are regular, so is  $A \cup B$ .

(The class of regular languages is closed under union)

Idea.

$\exists M_1$  accepts  $A$   
 $\exists M_2$  accepts  $B$   $\longrightarrow$  construct  $M_3$  accepts  $A \cup B$



$M_3$  accepts  $w$   
if at least one of  $M_1$  and  $M_2$   
accept  $w$ .

Proof:

$(\{0,1,a,b\})$

$\exists M_1 = (K_1, \Sigma, \delta_1, s_1, F_1)$  accepts  $A$

$\exists M_2 = (K_2, \Sigma, \delta_2, s_2, F_2)$  accepts  $B$

$M_3 = (K_3, \Sigma, \delta_3, s_3, F_3)$

$$\underline{K_3} = \underline{K_1 \times K_2}$$

$(q_1, q_2)$

$$\underline{s_3} = \underline{(s_1, s_2)}$$

$$\underline{F_3} = \{ (q_1, q_2) \in K_1 \times K_2 : q_1 \in F_1 \text{ or } q_2 \in F_2 \}$$

$$\delta_3 : K_3 \times \Sigma \rightarrow K_3$$

$$\delta_3((q_1, q_2), a) = (\delta_1(q_1, a), \delta_2(q_2, a))$$

for any  $q_1 \in K_1, q_2 \in K_2, a \in \Sigma$ .